



Database schema: MySQL 6m

Popular DBMSs available include MySQL, Oracle, PostgreSQL, Microsoft Access, MariaDB and dBASE, amongst others.

Among these, MySQL is widely used. The Stack Overflow Developer Survey placed MySQL as one of the most popular databases of 2024 (Stack Overflow, 2024).^{fnref:ade9c2} Most of the technical giants like Facebook, WordPress and X (formerly Twitter) utilise the MySQL DBMS.

This course uses MySQL as the DBMS for PHP programs and either phpMyAdmin (web application) or MySQL Workbench to manage the databases. Therefore, we will focus on MySQL here.

Why choose a MySQL schema?

Apart from its popularity, the reason we chose MySQL schema is that:

- Its platform is scalable and provides a secure way to manage the daily business transactions.
- Since it is also available as a cloud-based DBMS, it provides flexibility and reliability and is easy to integrate with most online tools.
- Many of the standalone xAMP (LAMP/WAMP/MAMP) tech stacks include MySQL as the default database.

Understanding MySQL schemas

Schemas are used by MySQL to define the database structure. They make use of indices to find a particular row in the entire table.

It is important that a schema is designed for particular queries with good logical and physical design because this is important to ensure you run high-performance queries. The schema objects are an important component in schemas; these include a trigger, a check constraint and a foreign key.

Managing MySQL schemas

Like with any database setup, you'll need a tool to take the logical schema (the conceptual design of the database and its relationships) and apply it to the actual database. This means the tool should allow you to edit SQL, manage tables, set up relationships, create indices, handle and query data and perform other tasks.

For MySQL, you could use various command-line tools to do this, but many people prefer a visual interface. The typical xAMP technology stack (such as MAMP, LAMP or WAMP) usually includes a database management tool. For MySQL, this tool is often phpMyAdmin, which we'll introduce next.

For more advanced, enterprise-level needs, you could use MySQL's own Workbench tool. However, for the purposes of this course, phpMyAdmin will be enough to handle your database management tasks.

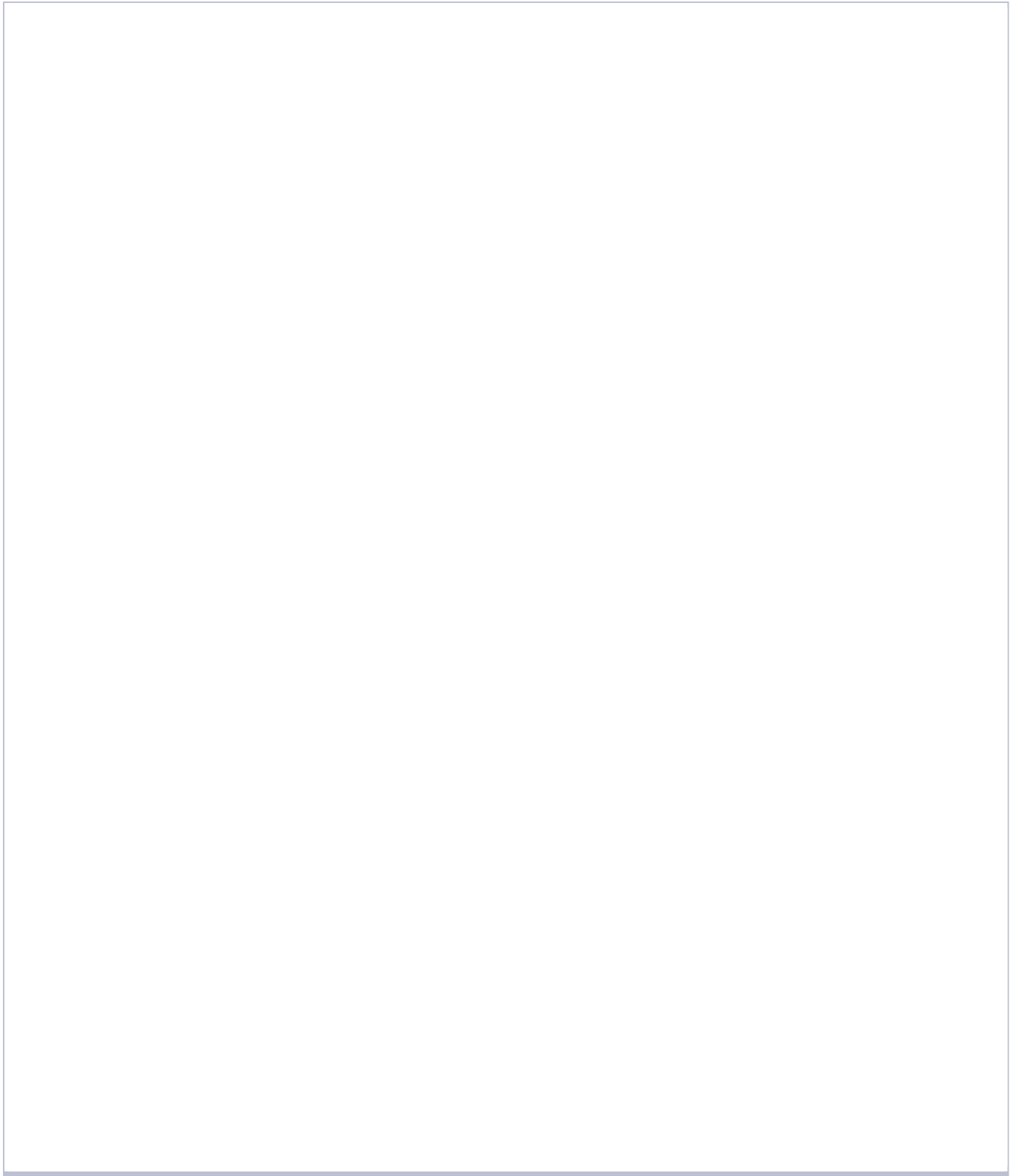
phpMyAdmin

Watch *Introduction to phpMyAdmin* (WordPress, 2024) (6:04)^{fnref:c4b961} to learn how WordPress users can navigate their site's database. This lesson explains the difference between site files and the database and guides users on how to access and explore it using phpMyAdmin. You'll also learn about key tables like wp_options and wp_posts and understand how data is structured.



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Design a database schema

Non-Assessed

What key elements would you include in a database schema to store information about a customer, and why?

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Next we'll create a MySQL database schema.



Thoughts on this page?

